TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of the Refractive Index of the Material of a Prism by a Spectrometer

Two angle measurements — the prism angle A and the angle of minimum deviation D_m — combine into μ = sin((A + D_m)/2) / sin(A/2), the most precise refractive index in the course.

PHY-1104 · PHYSICS-I SESSIONAL · EXPERIMENT 13 · LEVEL 1 TERM 1

1. Aim and theory

Aim: to find the refractive index of a glass prism. A ray through a prism of angle A is deviated by D, where D depends on the incidence angle: as incidence varies, D falls to a **minimum** D_m and rises again. At minimum deviation the ray passes symmetrically (incidence = emergence, and the ray inside is parallel to the base), and the geometry collapses into the exact formula

$$\mu = \sin(\,(A + D_m)/2\,) \,/\, \sin(\,A/2\,)$$

The spectrometer measures both angles by telescope positions on its divided circle: A from the reflections off the prism's two faces, D_m by tracking the refracted image while rotating the prism to the turning point of the deviation.

2. Apparatus and procedure

- **Apparatus:** spectrometer (collimator, prism table, telescope with vernier circle), glass prism, monochromatic source (sodium lamp) — μ is wavelength-dependent, so the light must be specified.
- Adjust the spectrometer: telescope for parallel rays (distant object / Schuster's method), collimator for a narrow bright slit, prism table levelled.
- **Prism angle A:** set the prism with its apex toward the collimator; catch the slit's reflection from each face in turn with the telescope; the angle between the two telescope positions is 2A.
- **Minimum deviation:** place the prism for refraction; find the spectral image; rotate the prism table slowly while following with the telescope — the image moves, halts, and reverses: that halt is minimum deviation. Read the telescope position; repeat with the prism removed (direct ray) — the difference is D_m.

- Compute μ. Report it with the wavelength (sodium yellow, 589.3 nm).

3. Worked calculation (model data)

> **RESULT**
>
> Reflection readings from the two faces differ by 120.0° → A = 60.0°.
> Direct-ray position and minimum-deviation position differ by D_m = 39.0°.
> μ = sin((60.0 + 39.0)/2)/sin(30.0) = sin(49.5°)/0.5000 = 0.7604/0.5000 = **1.521** — crown glass at the sodium wavelength, the standard teaching-prism result.

4. Precautions

- The minimum-deviation setting is the delicate one: approach the turning point slowly, bracket it (the image reverses direction there), and repeat the reading.
- Read both verniers of the circle and average — this cancels the eccentricity error of the divided circle.
- The prism must sit with its apex exactly over the table's centre and its faces vertical; re-level after any adjustment.
- Handle the prism by its rough (ground) faces only — fingerprints on the polished faces scatter the slit image.
- Use the same spectral line (sodium yellow) for D_m throughout; white light would smear D_m across colours (dispersion), which is a different experiment.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. What is special about the ray's path through the prism at minimum deviation?
2. For A = 60.0° and D_m = 52.0°, compute μ and name the likely glass.
3. Why is A measured by reflection as twice the angle between telescope positions?
4. Why must the light be monochromatic for this experiment, and what would a white-light source do to the D_m reading?
5. At minimum deviation, what is the relation between the angles of incidence and emergence, and between the internal ray and the prism's base?
6. A student's μ comes out as 1.52 for the blue line and 1.51 for the red line of the same prism. Is this an error? Explain.

Answer key

1. The path is symmetric: the ray enters and leaves at equal angles ($i_1 = i_2$) and runs parallel to the prism's base inside. Only then does the deviation reach its minimum — and only in that symmetric case does the simple μ formula hold.
2. μ = sin((60 + 52)/2)/sin 30 = sin 56°/0.5 = 0.8290/0.5 = 1.658 — a dense flint glass, as expected from the much larger D_m.
3. The telescope in each setting receives the beam reflected from one face of the prism. Because a reflected beam turns through twice the angle that the reflecting surface turns, the angle between the two telescope positions is twice the angle between the faces — the measured angle is 2A, and halving it gives A.
4. Because μ varies with wavelength (dispersion), "the" minimum deviation exists per colour. White light would give a spread spectrum whose every colour has its own turning point; no single D_m — and hence no single μ — could be read.
5. $i_1 = i_2$ (equal incidence and emergence), the internal refraction angles are equal too ($r_1 = r_2 = A/2$), and the internal ray is parallel to the base — the full symmetric configuration from which the formula is derived.
6. No — it is dispersion, and it is real physics: μ is genuinely higher for blue than for red in normal dispersion. The experiment's obligation is to report μ *with* its wavelength; quoting either value without the colour would be the error.
